UI/UX DESIGN

CHAPTER 1: FOUNDATIONS

1. Introduction to UI/UX

> "Design is not just what it looks like and feels like. Design is how it works." — Steve Jobs

What is UI (User Interface)?

- UI refers to the visual elements through which a user interacts with a digital product.
- Includes buttons, typography, layout, color schemes, icons, images, and responsiveness.
- Focuses on look, feel, and interactivity.

What is UX (User Experience)?

- UX is the process of creating products that provide meaningful and relevant experiences to users.
- Includes usability, accessibility, interaction design, and overall satisfaction.
- Focuses on problem-solving, user flows, empathy, and function.

Differences Between UI and UX:

Aspect	UI	UX
Focus	Visual and interaction elements	Overall user journey and experience
Involves	Color, typography, layout, animation	Research, wireframing, testing
Tools	Figma, Sketch, Adobe XD	Figma, Notion, Miro, Google Forms

Why UI/UX is Important:

- Enhances user satisfaction and engagement
- Increases accessibility and usability
- Essential for product success and business growth

Design Thinking Process

1. **Empathize** – Understand the users and their needs
2. **Define** – Clearly articulate the problem
3. **Ideate** – Brainstorm potential solutions
4. **Prototype** – Build rough versions of solutions
5. **Test** – Gather feedback and refine the solution

Career Roles in UI/UX:

- UX Designer
- UI Designer
- Product Designer
- UX Researcher
- Interaction Designer

Core Skills Needed:

- Empathy and problem-solving
- Visual design and layout skills
- Communication and collaboration
- Familiarity with design tools like Figma

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2. Understanding Users

> "If you want to understand how a lion hunts, don’t go to the zoo. Go to the jungle." — Jim Stengel

Who are your users?

- Individuals who interact with your product to solve a problem or achieve a goal.

User Personas:

- Fictional representations of target users based on real data.
- Include demographics, behaviors, needs, pain points, goals.
- Help guide design decisions with empathy.

Empathy Maps:

- Visual tool to understand what a user *says*, *thinks*, *does*, and *feels*.
- Divided into 4 quadrants + pains and gains.
- Encourages designing from a user's emotional perspective.

User Journey Mapping:

- A step-by-step visualization of a user’s experience when interacting with a product.
- Identifies touchpoints, emotions, roadblocks, and opportunities.
- Helps improve flow and experience.

Problem Statements:

- Clearly defined descriptions of user problems.
- Usually follow a structure like:
  - *[User]* needs a way to *[task]* because *[motivation]*.

Pain Points:

- Specific challenges or frustrations users face in the current process.
- Understanding pain points helps create better solutions.

Goals of Understanding Users:

- Build empathy
- Align product with user needs
- Prioritize features and UX improvements

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3. UX Research Basics

> "Research is formalized curiosity. It is poking and prying with a purpose." — Zora Neale Hurston

What is UX Research?

- The process of understanding user behaviors, needs, and motivations through observation and feedback.

- Helps make informed design decisions based on real data.

Types of UX Research:

1. Primary Research

- Collected directly from users.
- Examples: User interviews, surveys, usability testing, contextual inquiry.

2. Secondary Research

- Existing data and research from credible sources.
- Examples: Market analysis, analytics data, industry reports, competitor reviews.

Qualitative vs Quantitative:

- **Qualitative**: Non-numerical insights (e.g., user interviews, observations)
- **Quantitative**: Numerical data (e.g., survey results, heatmaps)

User Flows:

- Diagrams showing the complete path a user takes through a product to complete a task.
- Useful for identifying unnecessary steps or friction points.

Task Analysis:

- Breaking down the steps users take to complete tasks.
- Helps understand user goals and context.

Heuristic Evaluation:

- Expert review of interface using Jakob Nielsen’s 10 usability heuristics.
- Examples include:
  - Visibility of system status
  - Match between system and the real world
  - User control and freedom
  - Consistency and standards
  - Error prevention

When to Use Research:

- Early in the design process to identify problems
- During development to test design decisions
- Post-launch to measure success and iterate

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4. Wireframing

> "Wireframes are like the skeleton of your design—simple, sturdy, and essential." — Elfi

What is Wireframing?

- A wireframe is a low-fidelity, simplified visual representation of a layout or interface.
- Helps focus on structure and flow before visual design and content.

Low-Fidelity vs High-Fidelity Wireframes:

Aspect	Low-Fidelity	High-Fidelity
Detail Level	Basic layout, placeholder content	Real content, visuals, interactive look
Tools	Pen & paper, Figma wireframe kits	Figma (detailed with fonts/colors)
Usage	Early concept testing	Final layout preview before dev

Why Wireframe?

- Saves time and cost by catching issues early
- Focuses on layout and flow before aesthetics
- Encourages feedback and iteration

Wireframe Elements:

- Boxes for images, text blocks
- Placeholder buttons
- Layout structure: navigation, headers, footers

Best Practices:

- Keep it simple and minimal
- Use grayscale or neutral tones
- Label clearly (e.g., “CTA Button”, “Image Placeholder”)
- Use annotation if needed for explanation

Tools for Wireframing in Figma:

- Figma’s wireframe kit (community templates)
- Use Frames, Auto Layout, and Components
- Use grids for layout alignment

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5. Figma Essentials

> "Design begins with understanding your tools; mastery comes when they disappear into your flow." — Elfi

What is Figma?

- A browser-based UI/UX design tool for creating interfaces, wireframes, prototypes, and collaborative design systems.
- Offers real-time collaboration, version control, and plugins for advanced workflows.

Key Figma Features:

- **Frames**: Define boundaries for devices/screens (e.g., desktop, mobile)
- **Shapes & Vectors**: Use rectangle, ellipse, line, polygon tools for layout and icons
- **Text Tool**: Add and style text; supports fonts, spacing, hierarchy
- **Auto Layout**: Makes UI responsive and consistent with spacing
- **Components**: Reusable design elements (e.g., buttons, nav bars)
- **Variants**: Allow multiple states of a component (e.g., default, hover, clicked)
- **Assets Panel**: Library of reusable components and styles

Workflow Basics:

1. Create a **Frame** for your layout (e.g., iPhone 14)
2. Use **shapes** and **text** to build wireframe or mockup
3. Apply **Auto Layout** to ensure responsive resizing
4. Convert elements into **Components** for reusability

5. Organize using **Pages**, **Layers**, and **Naming Conventions**
6. Use **Prototyping Tab** to link screens with interactions
7. Share or Present directly from Figma

Figma Shortcuts (Popular Ones):

- `R` – Rectangle
- `T` – Text
- `A` – Frame
- `Shift + A` – Auto Layout
- `Ctrl/Cmd + D` – Duplicate
- `Ctrl/Cmd + G` – Group
- `Ctrl/Cmd + Alt + K` – Create Component

Best Practices for Beginners:

- Use proper naming for layers and components
- Utilize Auto Layout to reduce manual alignment
- Group similar elements
- Use grid systems for structure
- Keep files organized with pages and sections

Why Figma is Preferred:

- Cloud-based with autosave
- Team collaboration is seamless
- Works across platforms
- Large plugin ecosystem and free tier

CHAPTER 2: UI DESIGN PRINCIPLES

1. Visual Design Fundamentals

> “Design is intelligence made visible.” — Alina Wheeler

Elements of Visual Design:

- **Line:** Directs the user's gaze or creates boundaries
- **Shape:** Geometric (buttons), organic (illustrations), abstract (icons)
- **Color:** Communicates emotion, hierarchy, and branding
- **Typography:** Font choice, hierarchy, spacing, and readability
- **Space (White Space):** Breathing room that improves comprehension
- **Texture:** Adds realism or depth
- **Size & Scale:** Emphasizes importance and balance

Visual Hierarchy:

- Guides the user's eye through the design
- Achieved using size, contrast, alignment, repetition, proximity

Gestalt Principles:

- **Similarity:** Group similar elements together
- **Proximity:** Elements close to each other are related
- **Continuity:** The eye follows paths and curves
- **Closure:** The brain fills in missing information
- **Figure/Ground:** Separating foreground from background

Color Theory:

- **Primary, Secondary, Tertiary colors**
- **Analogous, Complementary, Triadic schemes**
- Contrast and accessibility (colorblind-friendly palettes)
- Use online tools like Coolors, Adobe Color

Typography Best Practices:

- Use 2–3 typefaces maximum
- Ensure readability across devices
- Pair fonts with contrast (e.g., Serif + Sans Serif)
- Set a clear hierarchy (Headings, Subheadings, Body)

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2. Layouts and Grids

> "Design is the silent ambassador of your brand." — Paul Rand

Grids in UI Design:

- Helps maintain alignment, structure, and rhythm
- Common types:
  - **Column Grid:** Divides page vertically for responsive design
  - **Baseline Grid:** Aligns text across sections
  - **Modular Grid:** For dashboards/cards

Layout Techniques:

- **F-pattern & Z-pattern:** Eye movement tracking for content scanning
- **Golden Ratio & Rule of Thirds:** Harmonious compositions
- **Visual Weight:** Use size, color, and placement to direct attention

Spacing & Padding:

- Consistent spacing = clean design
- Use 4/8/12 px spacing system (modular scale)

Responsive Design:

- Ensures UI looks good across screen sizes
- Consider mobile-first approach

Tools:

- Figma’s layout grids & auto-layout
- Plugins: Grid Generator, Grid Systemizer

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3. Typography in UI Design

> “Typography is the craft of endowing human language with a durable visual form.” — Robert Bringhurst

Role of Typography:

- Communicates tone and emotion
- Creates hierarchy and structure

- Enhances readability and user experience

Typographic Anatomy to Know:

- **Baseline:** The line on which most letters sit
- **X-height:** The height of lowercase letters
- **Ascender/Descender:** Parts of letters above/below x-height
- **Kerning:** Space between individual letters
- **Leading:** Vertical space between lines of text
- **Tracking:** Consistent spacing across entire word/paragraph

Font Types:

- **Serif:** Traditional, trustworthy (e.g., Times New Roman)
- **Sans Serif:** Modern, clean (e.g., Helvetica, Roboto)
- **Display Fonts:** Decorative, used for headings
- **Monospace:** Technical contexts (e.g., code editors)

Hierarchy Techniques:

- Size (e.g., H1 > H2 > Body)
- Weight (Bold, Regular, Light)
- Color and contrast

Best Practices:

- Stick to 1–2 typefaces
- Ensure contrast with background
- Use web-safe and readable fonts
- Responsive typography using rem/em

Tools & Plugins:

- Figma’s text styles
- Plugins: Google Fonts, Better Font Picker, Typescale

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4. UI Components and Design Patterns

> “Good design is obvious. Great design is transparent.” — Joe Sparano

UI Components (Building Blocks):

- **Buttons:** Primary, secondary, icon buttons
- **Forms & Inputs:** Text fields, radio buttons, checkboxes, dropdowns
- **Navigation:** Tabs, sidebars, menus, breadcrumbs
- **Cards:** Group related content or actions
- **Modals/Pop-ups:** Focused attention for tasks or alerts
- **Tooltips:** Contextual help/info
- **Loaders/Spinners:** Indicate background processes

Atomic Design Methodology:

- **Atoms:** Basic elements (buttons, inputs)
- **Molecules:** Groups of atoms (input + label + button)
- **Organisms:** Complex components (navbar, form section)
- **Templates:** Layouts with content placeholders
- **Pages:** Fully fleshed out screens

Design Patterns (UI Solutions):

- **Onboarding Flow**
- **Search and Filters**
- **Authentication/Login Flow**
- **Pagination vs Infinite Scroll**
- **Feedback & Alerts**

Best Practices:

- Consistency across all screens
- Reusability with Figma components and variants
- Follow platform-specific guidelines (Material Design, HIG for iOS)
- Use icon libraries and spacing tokens

Figma Features to Use:

- Components & Variants
- Auto Layout

- Interactive Prototypes
- Design Systems (Styles + Assets Library)

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5. Designing for Accessibility

> “Design is not just what it looks like and feels like. Design is how it works.” — Steve Jobs

Why Accessibility Matters:

- Inclusive design serves everyone
- Many users have temporary or permanent impairments
- Accessible design is ethical, legal, and expands reach

Accessibility Principles (POUR):

- **Perceivable:** Users must be able to perceive content (e.g., color contrast, alt text)
- **Operable:** UI must be usable with keyboard, screen readers
- **Understandable:** Clear labels, instructions, and feedback
- **Robust:** Works with assistive technologies across platforms

Color Contrast & Vision Impairments:

- Use WCAG guidelines (AA or AAA levels)
- Check contrast ratios (Figma plugins: Contrast, Stark)
- Avoid conveying info by color alone

Keyboard Navigation:

- Ensure all interactive elements are reachable and usable via Tab/Enter
- Maintain logical focus order

Screen Reader Compatibility:

- Use proper HTML semantics in code handoff
- Provide alt text for images, meaningful labels for inputs

Accessible Typography:

- Minimum font size (16px+)
- Good line height (1.4–1.6x font size)
- Avoid justified text (creates uneven spacing)

Figma Tools for Accessibility:

- Plugins: Stark, Able, A11y - Color Contrast Checker
- Use annotation layers to document accessibility features

Best Practices:

- Design with empathy
- Test with actual users and screen readers
- Prefer native elements for better accessibility support
- Maintain simplicity and clarity

CHAPTER 3: RESEARCH AND USER CENTERED DESIGN

> “If you don’t talk to your users, how do you know you’re designing the right thing?” — Erika Hall

User-Centered Design (UCD):

- An iterative design process that keeps the user at the center
- Involves understanding user needs, tasks, and feedback

Phases of UCD:

1. **Empathize:** User research, interviews, personas
2. **Define:** Identify pain points and opportunities
3. **Ideate:** Brainstorming and exploring possible solutions
4. **Prototype:** Creating wireframes and mockups
5. **Test:** Gathering feedback, iterating on design

UX Research Methods:

- **Qualitative:** Interviews, usability tests, field studies
- **Quantitative:** Surveys, analytics, A/B testing
- **Generative:** To discover problems and opportunities
- **Evaluative:** To test solutions and usability

Personas:

- Fictional profiles representing key user segments
- Based on research and real user behavior
- Includes: goals, frustrations, behaviors, tools used

Empathy Maps:

- Visual tool for understanding user mindset
- Sections: Think & Feel, See, Say & Do, Hear, Pains, Gains

User Journey Maps:

- Visualize the complete user experience across touchpoints
- Identify emotional highs and lows
- Helps find opportunities for improvement

Best Practices:

- Validate assumptions with real users
- Conduct research early and often
- Combine methods for holistic insight
- Document findings for team alignment

Wireframing & Prototyping in Figma

> “A prototype is worth a thousand meetings.” — IDEO

Wireframing:

- A visual guide that represents the skeletal framework of a digital product
- Focuses on structure, layout, and user flow (not visuals)
- Can be low-fidelity (sketches) or mid-fidelity (greyscale mockups)

Purpose of Wireframes:

- Plan page structure before visual design
- Identify key user interactions and information hierarchy
- Communicate ideas to stakeholders or developers

Types of Wireframes:

- **Low-Fidelity:** Quick, rough outlines; paper or digital (boxes, placeholders)
- **Mid-Fidelity:** Grayscale, includes layout, basic typography, icons
- **High-Fidelity:** Pixel-perfect mockups close to final UI

Wireframing Best Practices:

- Use grayscale tones to avoid distraction
- Start with mobile (mobile-first design)
- Focus on user tasks and goals
- Annotate for clarity

Tools:

- Figma (Frames, Auto Layout, Wireframe Kits)
- Balsamiq, Whimsical, Adobe XD

Prototyping in Figma:

- Simulates interaction flow between screens
- Helps test user journeys, animations, transitions

Key Concepts:

- **Hotspots:** Clickable areas
- **Connections:** Link frames with actions (e.g. “on click → navigate to”)
- **Smart Animate:** Smooth transition between states (e.g. toggle switch)
- **Overlays:** Modals, popups, dropdowns
- **Interactive Components:** Reusable elements with built-in states (e.g. buttons)

Prototype Testing Goals:

- Understand usability issues
- Gather user feedback early
- Reduce development rework

Tips:

- Keep interactions simple and intuitive
- Use prototypes to demonstrate UX to non-designers
- Test prototypes on real users for feedback

Deliverables:

- Interactive Figma prototype
- Documentation or notes for developer handoff

CHAPTER 4: ADVANCED FIGMA TECHNIQUES

1. Auto Layout Mastery

- Align and resize UI elements dynamically
- Stack elements vertically/horizontally
- Set spacing, padding, and alignment
- Responsive design adaptation

2. Components & Variants

- Reusable design elements (buttons, navbars)
- Variants for different states (hover, active, disabled)
- Update all instances from one source
- Use in design systems for consistency

3. Component Properties

- Add boolean toggles, text inputs, or instance swaps
- Customize components without detaching
- Example: Checkbox with toggled states

4. Design Systems in Figma

- Centralized library of components, styles, tokens
- Includes: Typography, Color Palette, Spacing, Icons, Grids
- Use `Assets` and `Team Library` features

5. Smart Animate & Motion

- Create smooth transitions (e.g. sliding menus, button states)
- Use same layers between frames
- Adjust easing and duration for natural feel

6. Interactive Components

- Embed interactions (hover/click) into components
- Enables real-time UI behavior in prototype mode

7. Team Collaboration

- Commenting, sharing, and version history
- Branching and merging designs (Pro feature)
- Effective handoff to developers with Inspect panel

8. Plugins & Widgets

- Use plugins like Iconify, Unsplash, Content Reel
- Boost productivity and design speed
- Explore community plugins for inspiration

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1. **Auto Layout Mastery**

> “Design systems love structure—and Auto Layout is the backbone.”

What is Auto Layout?
Auto Layout allows you to create dynamic and responsive designs in Figma. It helps in aligning, spacing, and resizing elements automatically as content changes.

Key Features:

- **Direction:** Stack elements *vertically* or *horizontally*
- **Spacing:** Define spacing between elements
- **Padding:** Space inside containers around content
- **Alignment:** Left, center, right, top, bottom
- **Resizing Behavior:**
  - **Fixed Width/Height** — stays constant
  - **Hug Contents** — fits tightly around content
  - **Fill Container** — expands to available space

Why Use Auto Layout?

- Saves time in resizing and spacing manually
- Ensures consistency across the design
- Enables responsive behavior (e.g., mobile to desktop)
- Works seamlessly with components and variants

Common Use Cases:

- Buttons that grow/shrink with text
- Navbars with evenly spaced menu items
- Cards that adapt to content length
- Lists or tables with dynamic rows

Pro Tips:

- Use `Shift + A` to apply Auto Layout
- Nest Auto Layouts (e.g. card inside a grid inside a frame)
- Rename layers properly for easy identification
- Test responsiveness by changing text or resizing parent frame

Example 1: Responsive Button

Goal: A button that adjusts width based on the text inside.

Steps:

1. Draw a rectangle.
2. Add text inside like “Click Me”.
3. Select both and press `Shift + A` to apply Auto Layout.

4. Set **padding**: e.g., 16px left/right, 8px top/bottom.
5. Set text layer to **Hug Contents**, and container to **Hug Contents**.
6. Now type a longer label — the button adjusts automatically!

Bonus: Add rounded corners and fill color for full styling.

Example 2: Card with Title, Description & Image

Goal: A card that holds an image on top and resizes text sections.

Steps:

1. Create a frame (Card container).
2. Add an image (rectangle or real image) at the top.
3. Below that, add:
   - A **title** text
   - A **description** text
4. Select all three (image, title, desc) → `Shift + A`.
5. Set direction to **Vertical**, spacing: 8–16px.
6. Card → Padding: 24px; Corner radius: 12px; Shadow if you like.
7. Set texts to **Fill Container** or **Hug Contents** depending on layout.

Use: Blog previews, profile cards, product tiles, etc.

Example 3: Navigation Bar

Goal: Horizontally spaced items that remain centered or justified.

Steps:

1. Add multiple text layers: Home | About | Projects | Contact
2. Select all → `Shift + A` → Set direction to **Horizontal**
3. Spacing: 32px; Alignment: Center
4. Wrap them in another Auto Layout frame (container)
5. Set outer container to **Fill Container** to stretch across screen width.

Pro Tip: Add a logo and a CTA (e.g., “Sign Up” button) using Auto Layout inside the same navbar frame.

Example 4: Dynamic List with Nested Auto Layout

Goal: Create a comment section where each row auto-adjusts.

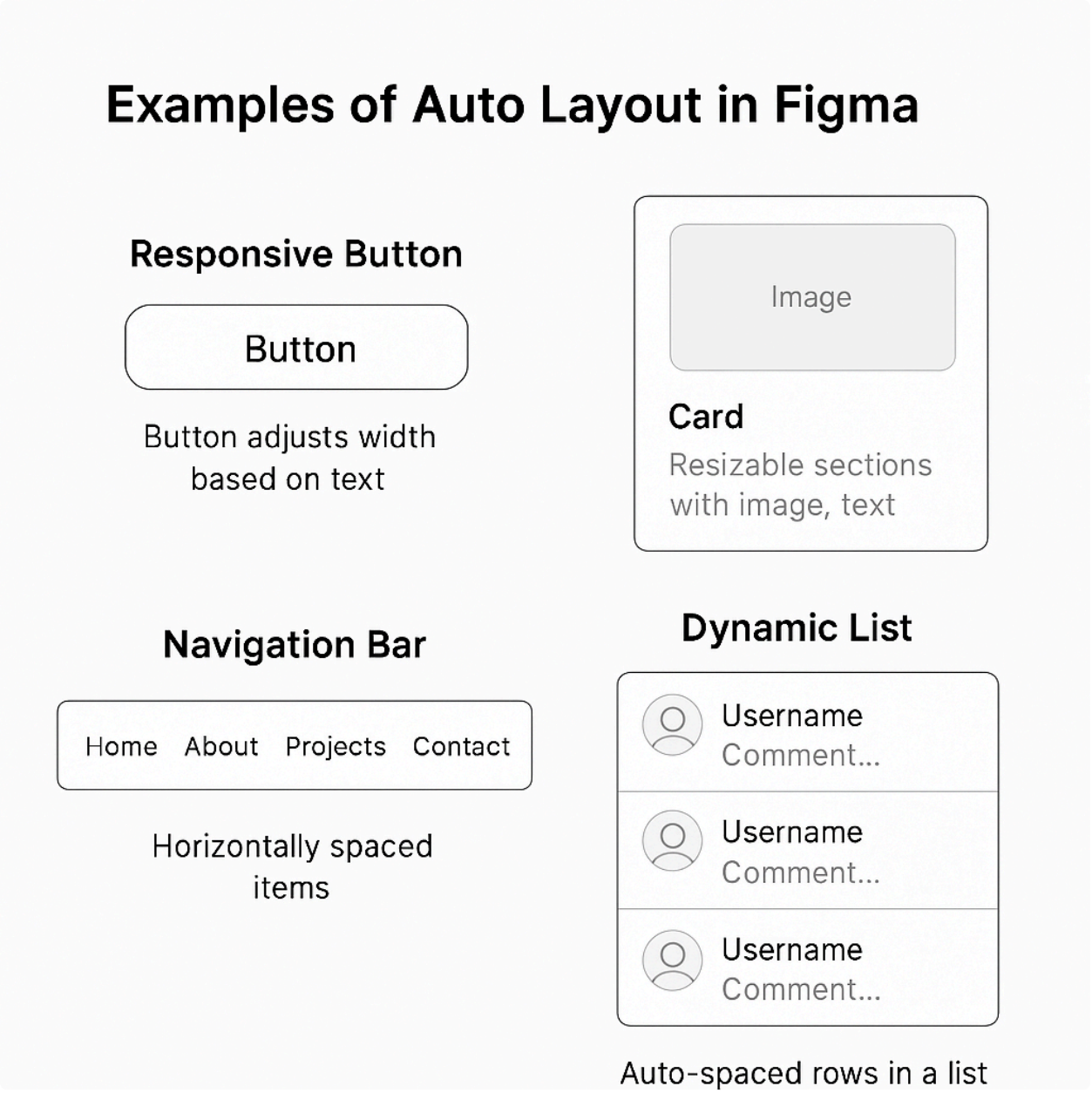
Steps:

1. Create a “Comment Row” frame: profile pic + username + comment.
2. Apply horizontal Auto Layout on that row.
3. Stack multiple rows vertically → Wrap in another Auto Layout (Vertical)
4. Set spacing between rows, padding, and background.

Outcome: Add/delete comments, and spacing stays perfect!

Use Case Summary:

UI Element	Auto Layout Direction	Special Behavior
Buttons	Horizontal	Hug contents + padding
Navbars	Horizontal	Fill container for stretching
Cards	Vertical	Image + Text stacking
Lists/Comments	Vertical (Nested)	Rows auto-spaced
Modals/Popups	Vertical	Centered & responsive



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2. Components and Varients

What are Components?

Components are reusable UI elements — think of them like templates. You create one “Master Component,” and reuse it throughout your design. Any change to the master updates all instances.

Example:

Let’s say you're designing a mobile app. You create a **Button Component** with the label “Submit.” You use this button on 12 different screens.

- Later, you change the font size from 14pt to 16pt in the master.

- Boom — all 12 buttons update instantly!

What are Variants?

Variants are *states or versions* of a single component grouped together — reducing clutter and making it easy to switch between styles (hover, pressed, disabled, etc.).

Example: Button Variants

Imagine this set of **Button Variants**:

Property	Values
State	Default, Hover, Pressed, Disabled
Size	Small, Medium, Large
Type	Primary, Secondary, Ghost

You can switch between them like dropdown options while using the component.

So, instead of creating:

- `Button/Default`
- `Button/Hover`
- `Button/Disabled`

You just have **one component** with multiple states as **Variants**.

Practical Use-Case

You’re building a form UI:

- **Input Field Component** with these variants:
  - State: Default, Focused, Error
  - Icon: Yes, No

This allows you to handle every case — all within a single component group.

Tips

- Use meaningful **property names** (like “state,” “type”)
- You can use **booleans** (e.g., “Show Icon: true/false”) for simple toggles
- Avoid too many nested components — keep things modular

Types of Component Properties:

Property Type	What It Does	Example Use Cases
Text	Lets you change the placeholder or label text	Buttons, headings, input labels
Boolean (true/false)	Toggles something on or off	Show/hide icon, show shadow, enable state
Instance Swap	Switch nested components inside	Swap an icon inside a button, different avatars
Variant Property	Allows user to switch between variants inside the component	Change from `Primary` to `Secondary` buttons

Real-World Example: Button Component

Component Setup:

- Text property: `label`
- Boolean property: `hasIcon`
- Instance Swap: `icon`
- Variant: `state` = `Default`, `Hover`, `Disabled`

Now in the instance:

- You can type any label: “Submit,” “Next,” “Continue”
- Toggle icon on/off
- Swap the icon to upload, arrow, checkmark
- Choose the state variant from dropdown

This means **you or your team never have to detach the component** — keeping everything scalable and clean!

Use-Case: Card Component

A **Product Card** could include:

- `Text`: Title and price
- `Boolean`: Show discount badge?
- `Instance swap`: Change product image or icon
- `Variants`: Available / Sold Out

All customizable from the right panel when using the card in your layout!

Tips for Mastery:

- Use **clear property names** (e.g., “Show Icon,” “Button Label”)
- Combine **Variants + Properties** for full flexibility
- Great for **design systems** and developer handoffs
- Reduces component clutter — one master, many faces!

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3. DESIGN SYSTEMS IN FIGMA

> “A design system isn’t just a collection of components; it’s a **language** for creating experiences.” — *Brad Frost*

What is a Design System?

A **Design System** is a centralized set of reusable **components**, **styles**, and **guidelines** that ensure **consistency and scalability** across all products and teams.

Think of it as your **digital architecture blueprint** — every button, color, spacing unit, and layout rule lives here.

Core Building Blocks:

Element	Description
Typography	Defined font styles: headings, paragraphs, captions
Color Palette	Primary, secondary, text, background, status colors (success, error)
Spacing Tokens	Defined units: 4px, 8px, 16px, etc. — ensures rhythm and alignment
Grids & Layouts	Columns, margins, and gutters — for structured, responsive design
Components	Reusable UI elements like buttons, inputs, cards
Variants	States of components: hover, active, disabled
Icons	Standardized icon sets (e.g., from Iconify)

How to Create a Design System in Figma:

1. **Create a New Figma File**

Name it `Design System` or `UI Kit`.

2. **Define Styles**:

- In the **Assets > Local Styles** panel, define:
  - Text Styles (H1, H2, Body, Caption)
  - Color Styles (Primary/Secondary, Background, Text, Error)
  - Effect Styles (Shadows, Blurs)

3. **Build Base Components**:

- Buttons (primary, secondary, disabled)
- Input Fields
- Icons
- Avatars
- Navbars
- Modals
- Cards

Use **Auto Layout** + **Variants** + **Component Properties** for max flexibility.

4. **Organize in Pages or Frames**:

- `Typography`
- `Colors`
- `Components`
- `Icons`
- `Grid & Layout`

Sharing Your Design System

- Publish components & styles using **Team Library**
- Allow others to **sync changes** across projects
- Use **naming conventions** for clarity (e.g., `btn/primary`, `input/filled`)
- Keep **documentation** (in FigJam, Notion, or inside a frame)

Bonus Tips:

- Make your system **responsive** — use Auto Layout, constraints, grids.
- Pair it with a **token system** (like Figma Tokens plugin) for dev integration.
- Always **document usage guidelines** — not just what to use, but *why*.

> "Animation can explain whatever the mind of man can conceive."
> — *Walt Disney*

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4. Smart Animate

Smart Animate is Figma's built-in animation feature that automatically animates **differences** between two frames — like position, size, color, opacity, or rotation.

You don't need to manually keyframe everything. Figma detects what changed and *smoothly transitions* it.

How it Works:

1. You create **two frames** (Frame A → Frame B).
2. Add an **interaction** (e.g., "On click → Navigate to Frame B").
3. Choose **Smart Animate** as the animation.
4. Figma compares layers with **same names** and animates property changes.

What Can It Animate?

Property	Smart Animate Support
Position (X, Y)	✅
Size (W, H)	✅
Rotation	✅
Color	✅
Opacity	✅
Radius	✅
Stroke Width	✅
Text (limited)	⚠️ Only font style/size, not content
Visibility	✅ Works via Opacity
Component State	✅ With Variants

Example:

Imagine a card that **expands** on click:

Frame 1:

- Card is small
- Image is thumbnail
- Button is hidden (opacity 0)

Frame 2:

- Card is full-screen
- Image is larger
- Button fades in (opacity 100)

Add interaction: “On click → Navigate to Frame 2 → Smart Animate (300ms, ease-in-out)”

Result: Smooth expansion and fade-in — no coding needed!

With Variants

Smart Animate works *between variants* too!

E.g., A button with states:

- Default
- Hover
- Pressed

Use `While Hovering → Change to Hover → Smart Animate (150ms)`

Best Practices:

✅ **Name your layers consistently**
Smart Animate only works if Figma can *match* layers. Rename like:

- `img-hero`
- `btn-login`

✅ **Use Auto Layout + Constraints**
Makes resizing and transitions much smoother.

✅ **Keep animations short**
200–400ms is usually enough. Too slow feels laggy.

✅ **Use Easing functions**

- Ease-in: Feels like acceleration
- Ease-out: Decelerates, feels more natural
- Ease-in-out: Smooth both ways

BONUS: Animation Types in Figma

Transition Type	Description
Instant	No animation
Dissolve	Smooth fade between screens
Move In/Out	Slide in/out from a side
Push	Slides old frame away as new comes in
Slide In/Out	Like Move, but _overlays
Smart Animate	Smooth Interpolation of changes

---

5. INTERACTIVE COMPONENTS IN FIGMA

> "Design is not just what it looks like and feels like. Design is how it works."
> — *Steve Jobs*

What Are Interactive Components?

Interactive Components allow you to embed **interactions *inside* a component** — so its hover, click, toggle, etc., work **everywhere** the component is used, without needing multiple frames or prototype connections.

Think: *Reusable buttons that know how to animate when hovered or clicked — built into the component itself.*

Why They’re Game-Changing

Without Interactive Components	With Interactive Components
You duplicate frames to show states	Use one component with states
Need connections between all states	Figma auto-handles transitions
Messy prototypes with lots of nodes	Clean, minimal, fast

Example: Button Component

Step 1: Create a Component with Variants

- Variant 1: `Default`
- Variant 2: `Hover`
- Variant 3: `Pressed`

Step 2: Add Interactions Inside the Component Set

- `While Hovering → Change to Hover → Smart Animate`
- `While Pressing → Change to Pressed → Smart Animate`
- `Mouse Leave → Change to Default`

Step 3: Use the Component Anywhere

Now, *every instance* of this button will **behave interactively** in the prototype — with no need to connect multiple frames!

Use-Cases for Interactive Components:

Component	Common States
Buttons	Default, Hover, Pressed, Disabled
Toggles/Switches	On, Off, Hover
Tabs	Active Tab / Inactive Tabs
Accordions	Collapsed / Expanded
Tooltips	Show on hover
Navigation Menus	Open/Close states

Component	Common States
Checkboxes	Checked, Unchecked, Indeterminate

Best Practices

✅ **Name your variants properly**: `Default`, `Hover`, `Pressed` — makes maintenance easy
✅ Use **Smart Animate** for beautiful transitions
✅ Keep **interactions limited** inside components — don't overcomplicate
✅ Combine with **Component Properties** for maximum flexibility
✅ Test thoroughly in **Prototype mode**

Limitations (as of now)

- Some **nested components** with interactivity don't behave perfectly
- Doesn't support **external navigation** (i.e., can't go to another frame from inside component)

Pro Tip:

Want a **toggle switch** that animates on click?

1. Create 2 variants: `On` and `Off`
2. Inside the component set, do:
   - `On Click → Change to (Other State) → Smart Animate`
3. Done! It'll toggle perfectly **in a single frame**.

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6. TEAM COLLABORATION IN FIGMA

> "Alone we can do so little; together we can do so much."
> — *Helen Keller*

What Makes Figma Perfect for Teams?

Figma is **cloud-based**, which means:

- No downloads
- No file conflicts

- **Everyone works on the same file in real time**
- Instant feedback, edits, and progress

Real-Time Collaboration Features:

Feature	What It Enables
Multiplayer Editing	Multiple teammates can design together — live!
Live Cursors	See who’s doing what, in real time
Comment Mode	Leave comments anywhere on the canvas
Version History	Track changes and revert if needed
File Sharing	Share via link, control permissions (edit, view, comment)
Slack / Email Notifs	Get notified when tagged or files are updated

Pro Tips for Healthy Collaboration:

- Use **Sticky Notes** or **Widgets** in FigJam to ideate together
- Schedule **design reviews** within Figma using comment mode
- Avoid "dirty drafts" — keep personal exploration in separate pages
- Use **versioning** before major edits
- Respect shared files — don’t overwrite unless aligned

Figma for Cross-Functional Teams:

Role	How They Use Figma
Designers	UI/UX design, prototyping, feedback loops
Developers	Specs, component inspection, commenting on logic
PMs	Flow approvals, wireframes, feature suggestions
Marketing	Landing page mockups, social posts, branding
Clients / Stakeholders	Reviewing progress, approving final screens

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7. PLUGINS & WIDGETS IN FIGMA

> *"Design is intelligence made visible — and plugins are the brain boosters."*

What Are Plugins?

Plugins are third-party tools that help automate, simplify, or extend Figma's functionality.
They **only affect your own file view** and are **not seen by others unless they use it too.**

Examples:

- Generating content
- Automating tasks
- Importing data
- Cleaning up design files

Installing & Using Plugins

1. Go to **Main Menu → Plugins → Browse Plugins in Community**
2. Search → Click on a plugin → Install
3. Use via **Right Click → Plugins → [Plugin Name]**

Pro Tip: You can "star" plugins and access them quickly later!

Must-Have Plugins for UI/UX Designers:

Plugin	What It Does
Content Reel	Adds realistic content (names, emails, images)
Icons8 / Iconify	Access 100k+ icons from within Figma!
Unsplash / LoremPicsum	Add random or themed stock images
Google Sheets Sync	Pull real data into your mockups from a sheet
Autoflow	Easily create user flows between frames
Able	Check color contrast for accessibility compliance
Image Palette	Extracts color palette from images
Remove BG	One-click background removal from images
Similayer	Select similar objects based on shared properties
Design Lint	Spot inconsistencies in spacing, colors, alignment

What Are Widgets?

Widgets are collaborative tools used *in FigJam* (Figma's whiteboard tool).
Think: Sticky notes, voting tools, charts, games!

They **help with brainstorming, team exercises, feedback sessions**, etc.

Popular Widgets for Team Workflows:

Widget	Purpose
Sticky Notes	Brainstorming ideas collaboratively
Voting / Emoji Reactions	For quick decision making or team feedback
Persona Generator	Quickly create empathy maps and personas
Timer	Great for timeboxing discussions or workshops
Story Map / Journey Map	Use during product planning or sprint reviews

Elfi's Tips:

- Use **Plugins** during **design phase** to streamline UI work
- Use **Widgets** during **planning / ideation** sessions in FigJam
- Combine both for a smooth **end-to-end design process**

Accessing Plugins & Widgets from Community:

- Go to the **Community tab** in Figma (left panel)
- Browse → Install → Use in your file
- You can even **build your own plugin or widget** using Figma's Plugin API! (JavaScript-based)

CHAPTER 5: CASE STUDIES, PROJECTS AND PORTFOLIOS

> *"Your portfolio is your passport. Make it speak for you before you even enter the room."*

1. **What is a Case Study in UI/UX?**

A **case study** is a storytelling tool that shows:

- Your **process** (not just final screens!)
- Your **thinking** and **problem-solving skills**
- How you handled **real-world constraints**

It usually answers:

- What problem were you solving?
- Who were you designing for?
- What steps did you take?
- What was the result?

Structure of a Case Study

Section	Description
Overview	Brief summary of the project (goal, platform, users)
Problem Statement	What were the pain points or user problems?
User Research	Methods used (surveys, interviews, personas, etc.)
Ideation	Sketches, wireframes, crazy 8s, brainstorming
Design Process	Low–hi fidelity mockups, iterations, feedback
Prototyping	How interactions were planned/tested
Testing & Feedback	Usability testing, what worked/didn't
Outcome & Learnings	Final design and what you learned

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2. **Building a UI/UX Project**

Here's a sample process you could follow:

> Project Idea: A Sustainable Shopping App
> Goal: Help users find eco-friendly products with verified green labels.

Steps:

1. Define Problem: "Users struggle to find truly sustainable products online."
2. Research: Interviews with eco-conscious shoppers

3. Persona: Riya, 22, college student, values ethical brands
4. Flow Diagram: From product search → cart → checkout
5. Wireframe: Layout for product card, filters, checkout
6. UI Design: Use green tones, clean layout, icons
7. Prototype: Use Figma interactions, smart animate
8. Test: Ask 3–5 people to test and give feedback
9. Refine: Based on real feedback
10. Document Case Study!

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3. **Portfolio Building Tips**

Platform Options:

- Behance
- Notion
- Dribbble
- [GitHub Pages] or [Personal Website (recommended!)]

Tips:

- Add *3–5 solid case studies*
- Explain your role clearly (especially in group projects)
- Include *process images* not just final screens
- Be honest about challenges and failures too

🎥 Bonus: Create a **screen recording** of your prototype for visual appeal

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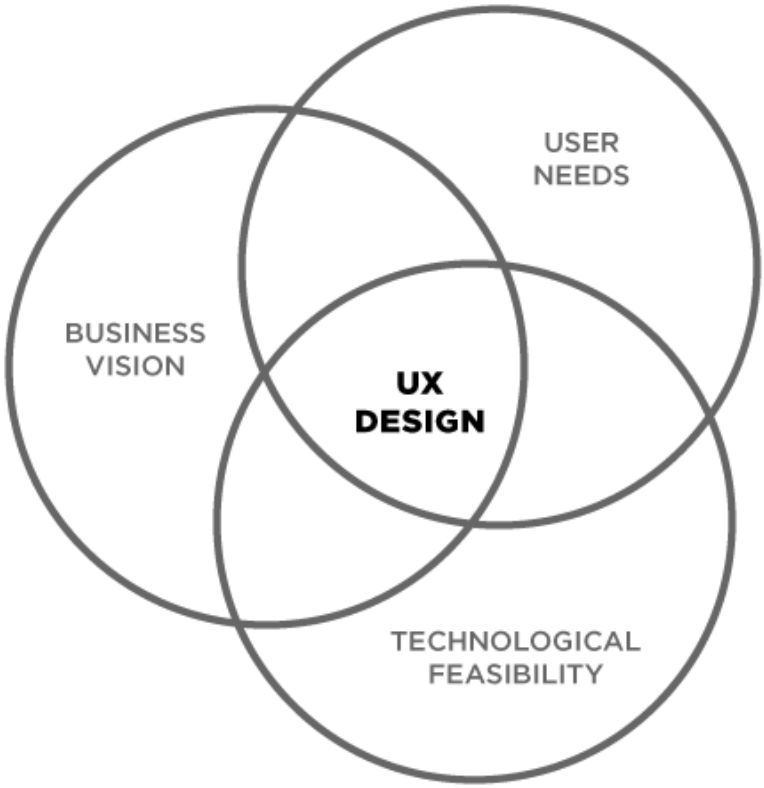
4. What Makes a Good Project?

✅ Solves a **real problem**
✅ Shows clear **research-to-design process**
✅ Involves **feedback loops** and **iteration**
✅ Ends with **takeaways** and future scope

UI/UX BASICS

What is UX?

- **UX** stands for **user experience** or **user experience design**. It's also sometimes abbreviated as **UXD, UED or XD**.
- **UX** is the naked experience of a product: how a product functions, not how it looks.
- **UX Design** is the process of enhancing user satisfaction of a product through increased usability, accessibility, and pleasure provided in the interaction with the product. User experience design encompasses note only traditional human–computer interaction design, but also all aspects of a product or service as perceived by users.

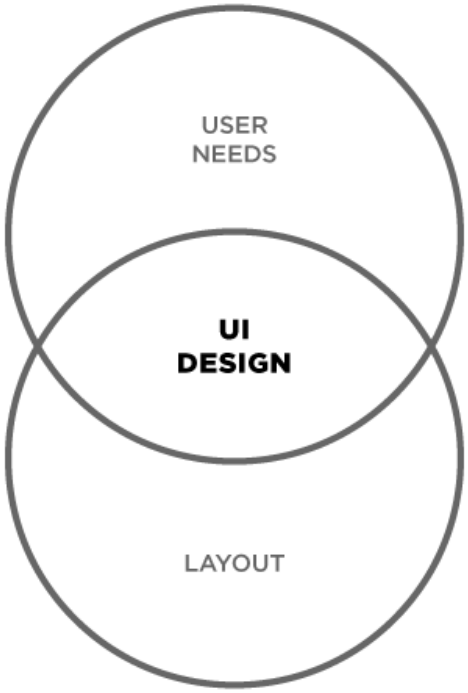


UX design is the merger of user needs, business vision and technological feasibility. The result of UX design is a set of low-resolution wireframes (a basic visual guide for how a product will function) that are deeply connected with user research.

UI/UX BASICS

What is UI?

- **UI** stands for **User Interface** or **User Interface Design**. It's also sometimes known as **user interface engineering**.
- **UI** is the design of user interfaces for machines: how a product looks and feels, not how it functions.
- **UI Design** is the process of making the user's interaction as simple and efficient as possible, in terms of accomplishing their goals (also known as user-centered design).



UI design is the merger of user needs and visual design. The result of UI design is a set of high-resolution wireframes (a visual representation of a product).

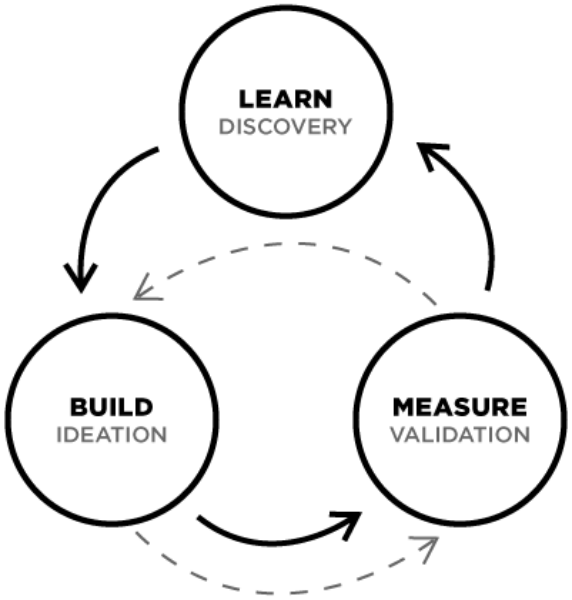
UX and UI Are Not the Same

UX is not UI. UI is not UX. Let's say our product is a home. UX designers will create the architectural plans for the house and decide where to build it. UI designers will concentrate on landscaping, what color the house will be, and other visual decisions.

UI/UX BASICS

What is the process of UX?

UX has 3 key phases: **discovery, ideation,** and **validation**. However, UX design is cyclical and you will often need to repeat certain steps and even the entire process multiple times.





The final outcome will be a set of low-resolution wireframes: a draft of the function and structure of a product.

UI/UX BASICS

1. Discover

UX always starts with **discovery** — interviewing potential customers to understand what the target audience needs and talking to stakeholders to understand their goals and competitive analysis. Discovery is all about WHY - xxx. During discovery you will validate your problem (your product is there to solve that problem), identify your end users, and determine project goals.

Methods:

- User Research (Interviews, Ethnography)
- Empathy Mapping
- Task Analysis
- Stakeholder Mapping
- Service Blueprints
- Analytics and Heuristics
- Competitive Analysis

Outcome: Problem Validation, User Personas, Project Goals

2. Ideate

Next comes **ideation** — using a variety of tools to imagine a solution that solves the user problem, while aligning with the company goals within technological possibility. Ideation is the process of finding out HOW. How will you create a solution that solves the users deepest needs in a delightful manner? Designers will implement a variety of tools to figure out how to solve the user problems. This process is very much like a funnel, where the solution is very wide at the beginning, and the goal of the process is to quickly, envision and test products with target customers in order to pivot and define. During ideation you will organize your discovery, explore options, and develop wireframes and prototypes.

Methods:

- Sketching
- Wireframes
- Information Architecture
- Journey Mapping / Page Flows
- User Journey Writing
- Paper Prototypes
- Interaction Design

Outcome: Solution Exploration

Notes: _____

_____

_____

UI/UX BASICS

3. Validate (Test, Prototype)

The UX process ends with **validation** — the testing of wireframes and prototypes to iterate on the interface until it's intuitive and delightful. Validation is when we finally know WHAT we're building. During this phase, designers will create wireframes or prototypes, and test them with users during a process called usability testing to evaluate how an actual user will react to the product. The designer observes, asks open-ended questions and iterates on the wireframes based on this feedback. Validation testing is giving those wireframes or prototypes to real users. You're tracking actual interaction with the prototype here as well as confirming previously held assumptions. The results of validation testing should be changes in flow and layout, though likely not scope anymore. During validation you will validate your ideas, learn, and plan for the next iteration.

Methods:

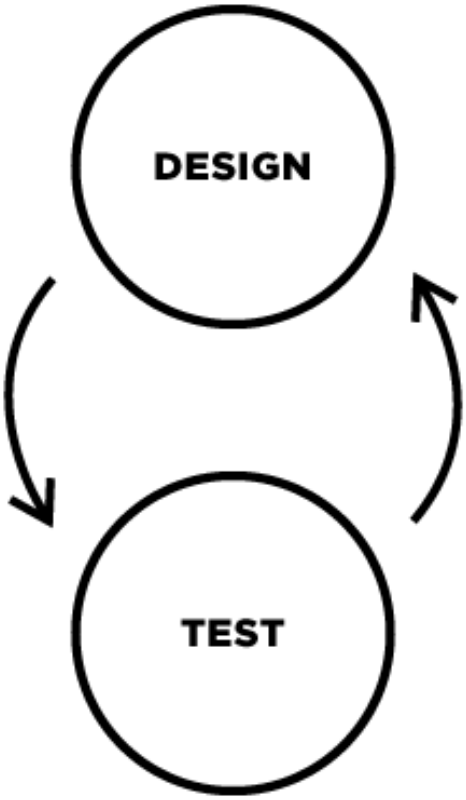
- Accessibility
- Usability Testing
- Feedback integration
- Interactive Design
- Retrospectives
- Release

Outcome: Solution Scalability, Low-Res Wireframes

What is the process of UI?

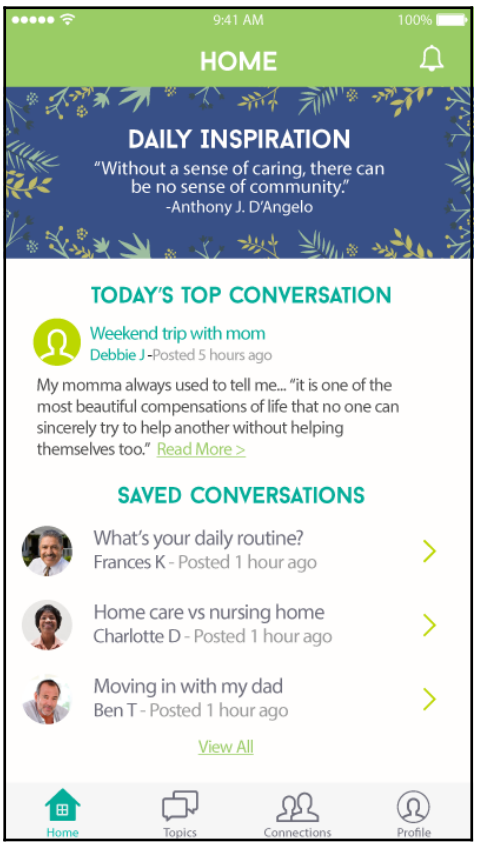
UX has e key phases: design and testing. UI design precedes the development of the product's functional elements.

UI design is the creation of the finished interface; its focus is on the visual and emotional feel of the product. UI design establishes the layout, colors, typography and interactivity to visually communicate the flow of the screens in an intuitive manner. The UI design process must balance technical functionality and visual elements to create a system that is not only operational but also usable and adaptable to changing user needs.

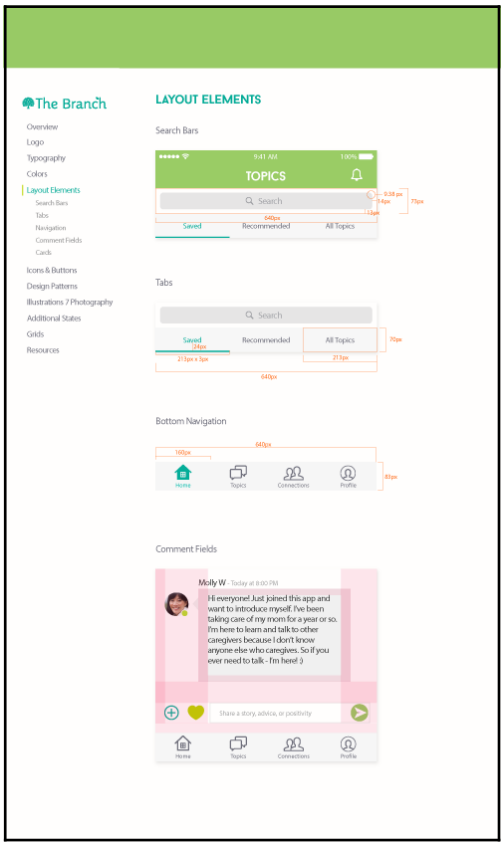


UI/UX BASICS

The final outcome will be a set of high-resolution wireframes (a prototype or draft version of a product) and a style guide.



Page from high-resolution wireframes



Page from a style guide

1. Design

After ideation is design — this puts your ideas to the test. During design you will commit to internally validated ideas and test those ideas with users. After you complete the design process, you will likely bring what you've learned back to step 2 (ideation) and repeat this step again. This can happen multiple times.

Methods:

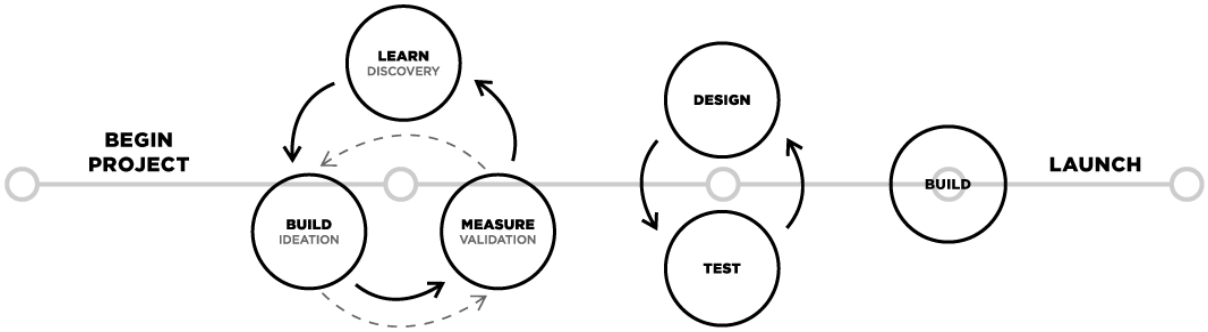
- Design Sprints
- Style Guides

2. Validate (Test, Prototype)

After you design, you will test to validate your design ideas. During testing you will test your ideas with users. After you complete the testing process, you will likely bring what you've learned back to step 1 (design) and repeat this step again. This can happen multiple times.

UI/UX BASICS

How Do UI and UX Fit Together?



The processes of UX Design and UI Design are flexible and there is no "right" way of bringing them together for one project. Typically, a project will need UX Design first and then UI Design. However, for existing products, either process may be used alone to improve either UX or UI.

Practical Applications and Design Thinking

UX and UI are driven by Design Thinking, which refers to creative strategies designers use during the process of designing. This approach is also useful to resolve issues outside of professional design practice, such as in business, social, or personal contexts.

There are 5 keys to design thinking, and you can apply them to your everyday life.

1. **Empathize**—When you design, you're not primarily doing it for yourself. You're doing it with other people's needs and desires in mind. Focus on the person or problem that you are serving. Find ways to serve them better. Help their loves be better each day. Empathize first.
2. **Discover**—Try to narrow down your problem to the root cause. Jump in with an open mind, without criticism or opinions. Seek to understand before you are understood.
3. **Ideate**– Be imaginative, create ideas to solve the problem you identified in step 2.
4. **Prototype**– Place the concept of being perfect aside. Embrace failure to master the process. Don't be cautious.
5. **Test**—Cultivate self-awareness by asking yourself, "what do I really want to see or experience?" Never limit yourself.

Resources:

- *UX Strategy: How to Devise Innovative Digital Products that People Want* by Jaime Levy
- *Designing Interfaces* by Jenifer Tidwell
- *Sprint: How to Solve Big Problems and Test New Ideas in Just Five Days* by Braden Kowitz and Jake Knapp
- *The Achievement Habit: Stop Wishing, Start Doing, and Take Command of Your Life* by Bernard Roth